

# CH365 Chemical Engineering Thermodynamics

## Lesson 17

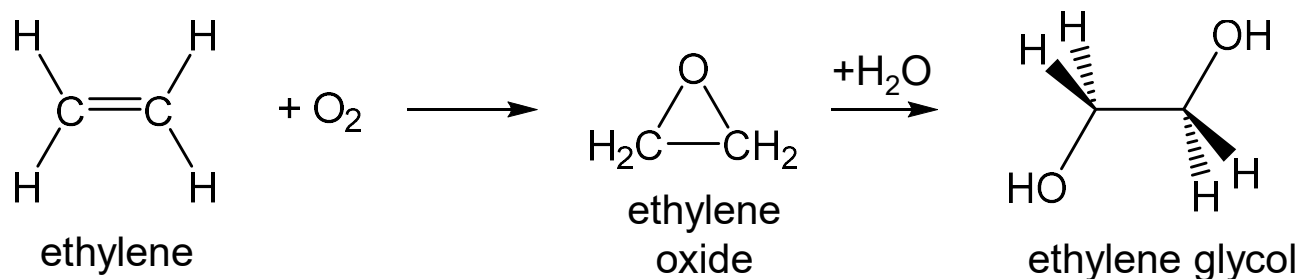
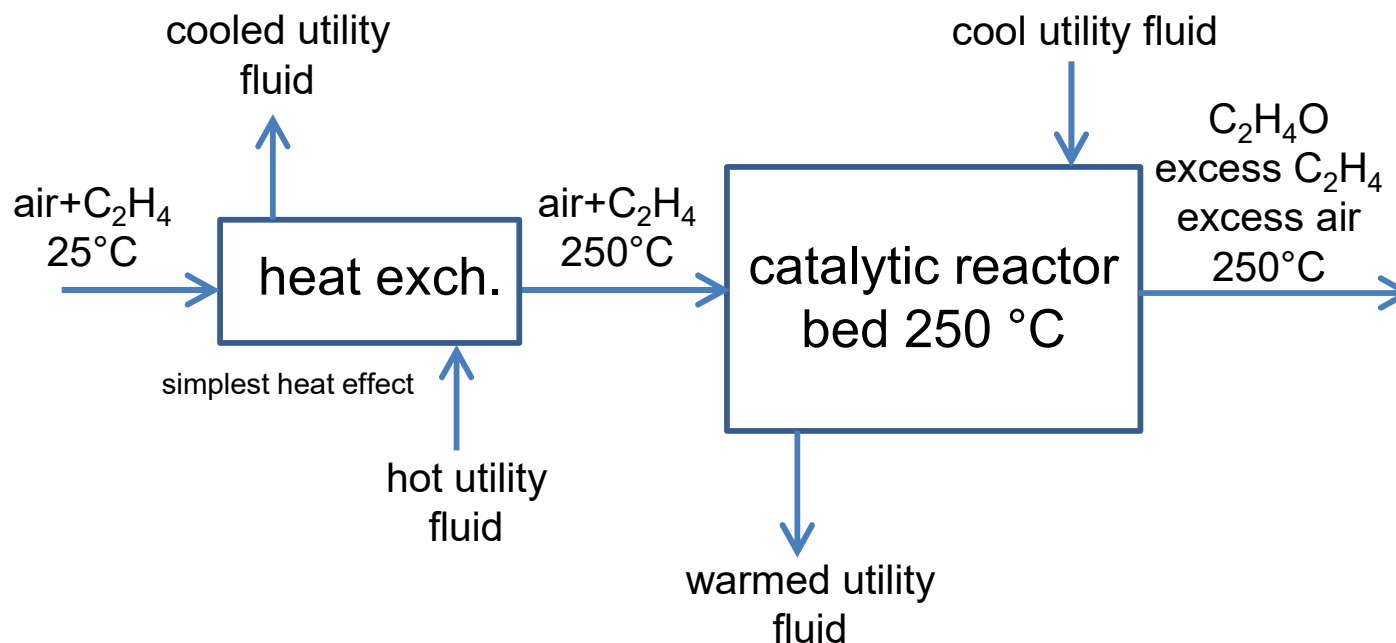
Latent Heat and Heats of Reaction,  
Formation, and Combustion

# Ethylene Glycol Process

Slide 2

Sensible heat effects – temperature change

➡ Latent heat effects – no temperature change



# Latent Heats of Pure Substances

One component evaporating or condensing – No temperature change but heat is transferred. Latent heat of fusion or vaporization.

# Latent Heats of Pure Substances

One component evaporating or condensing.

# Latent Heats of Pure Substances

One component evaporating or condensing.

# Standard Heat of Reaction

Heat effects in chemical reactions are manifestations of the differences in molecular structure.

There are a large number of chemical reactions, each of which can be carried out at an essentially infinite set of process conditions.

Tabulation of all possible heat effects for all possible reactions is impossible.

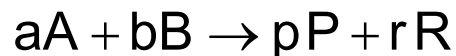
Heats are calculated from data for reactions conducted under standard conditions.

Heat associated with a chemical reaction depends on the temperatures of the reactants and products. A consistent standard treatment is that products and reactants are at the same temperature.

# Standard Heat of Reaction

Heats are calculated from data for reactions conducted under standard conditions.

Products and reactants are at the same temperature



Standard Heat of Reaction is the enthalpy change when  $a$  moles of  $A$  and  $b$  moles of  $B$  in their standard states at temperature  $T$  react to form  $p$  moles of  $P$  and  $r$  moles of  $R$  in their standard states at temperature  $T$ .

Standard State Definitions:

Gases: ideal gas at 1 bar

Liquids and Solids: real pure liquid or solid at 1 bar

Degree symbol designates standard state

$C_P^\circ = C_P^{\text{ig}}$  for gases

Subscript “f” designates “formation.”

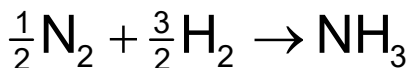
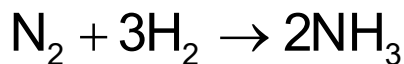


Table C.4, page 672

$$\Delta H_{f,298}^\circ = -46,110 \text{ J}$$



$$\Delta H_{298}^\circ = -92,220 \text{ J}$$

$\Delta H_{298}^\circ$

Standard value at 298.15 K

Standard state properties are functions of  $T$

# Standard Heat of Formation

In a *formation reaction*, a compound forms from its constituent elements.

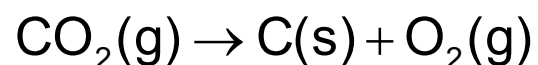


The standard-state heat of formation is based on formation of one mole of a substance from its constituent elements in their standard states and is designated as  $\Delta H_{\text{f},298}^{\circ}$ .

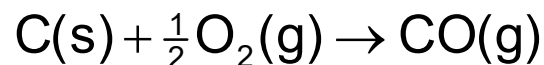
The standard heat of reaction can be calculated if the heats of formation of the reactants and products are known.

Water-gas-shift reaction:

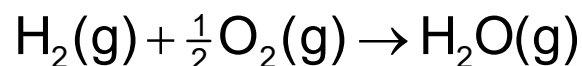
Table C.4, page 672



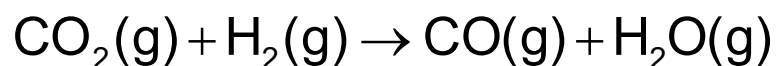
$$\Delta H_{298}^{\circ} = 393,509 \text{ J} \quad \text{negative in Table C.4 (listed as product)}$$



$$\Delta H_{298}^{\circ} = -110,525 \text{ J}$$



$$\Delta H_{298}^{\circ} = -241,818 \text{ J}$$



$$\Delta H_{298}^{\circ} = 41,166 \text{ J}$$

# Questions?